

EXPLORATORY DATA ANALYSIS — CARS DATASET

Cleaning • Exploration • Visualization • Insights

GADEELA AKSHAYA

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Dataset source: Provided Cars dataset (CSV supplied for the project)

01 | PROJECT OBJECTIVE

Understand the structure, quality and patterns in used-car data.

Goal Clean, explore and explain meaningful patterns in automobile data.

Dataset 120 cleaned records • 15 original columns + derived car_age

Questions Which brands/fuels are common? How are prices distributed? How do age, mileage and transmission relate to price?

Approach Inspect → Clean → Describe → Visualize → Interpret

02 | DATA QUALITY & CLEANING

Before: 123 rows × 15 columns → After: 120 rows × 16 columns

Standardized labels	Brand, fuel type, transmission, owner type, city and insurance labels were normalized.
Corrected types	Year and kilometres driven were converted to numeric values.
Invalid values	Negative kilometres were treated as missing before median imputation.
Missing values	Missing brand/transmission filled with mode; numeric missing values filled with median.
Duplicates	Duplicate rows were removed; final duplicate count = 0.

Note: listed_date was retained as supplied; no unsupported transformation is reported.

03 | DESCRIPTIVE ANALYSIS

Key summary statistics from the cleaned dataset

RECORDS

120

MOST COMMON BRAND

Maruti — 35 cars

MINIMUM PRICE

■189,000

AVG. AUTOMATIC PRICE

■840,506.85

BRANDS

9

MOST COMMON FUEL

Petrol — 52 cars

MAXIMUM PRICE

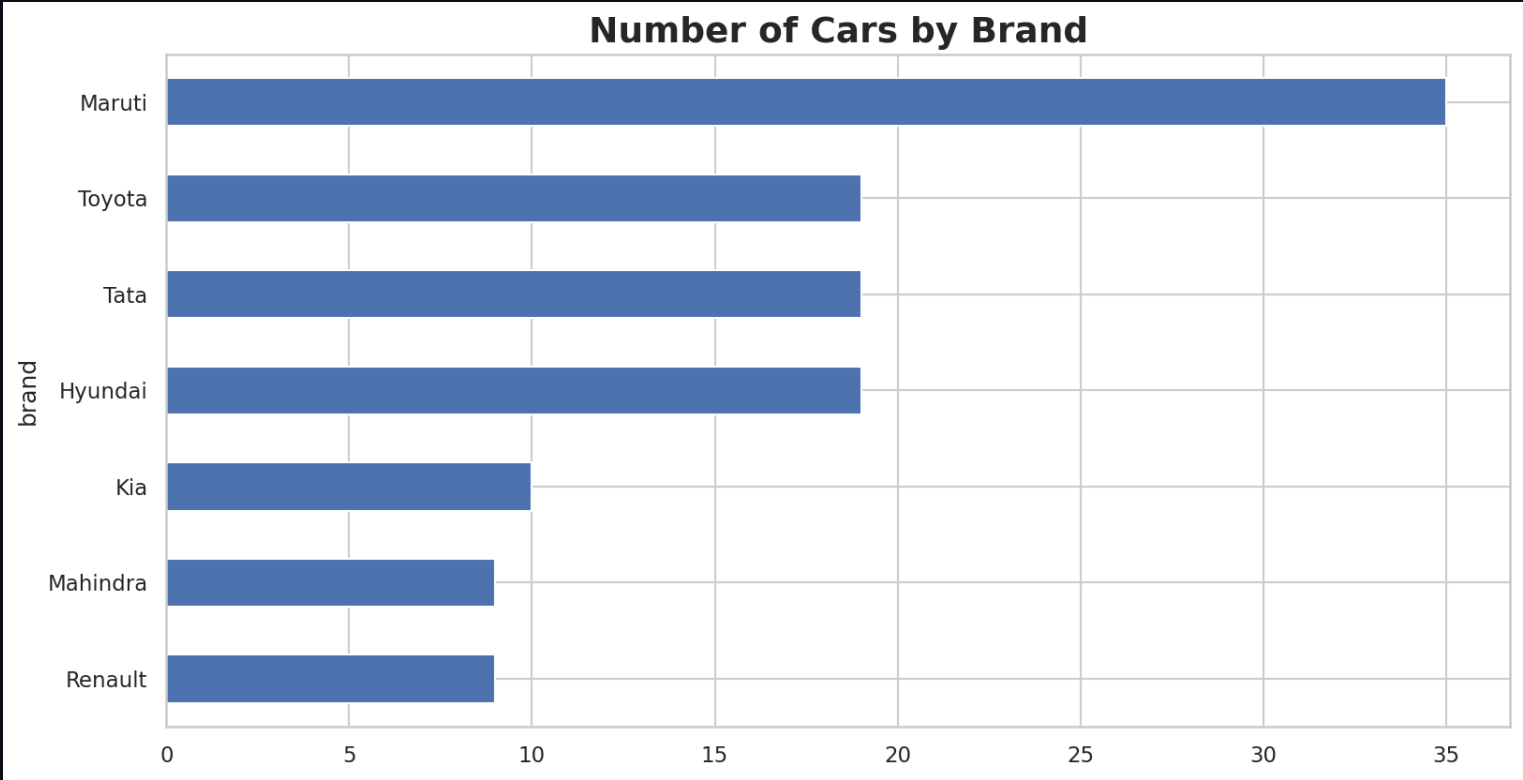
■9,500,000

AVG. MANUAL PRICE

■733,191.49

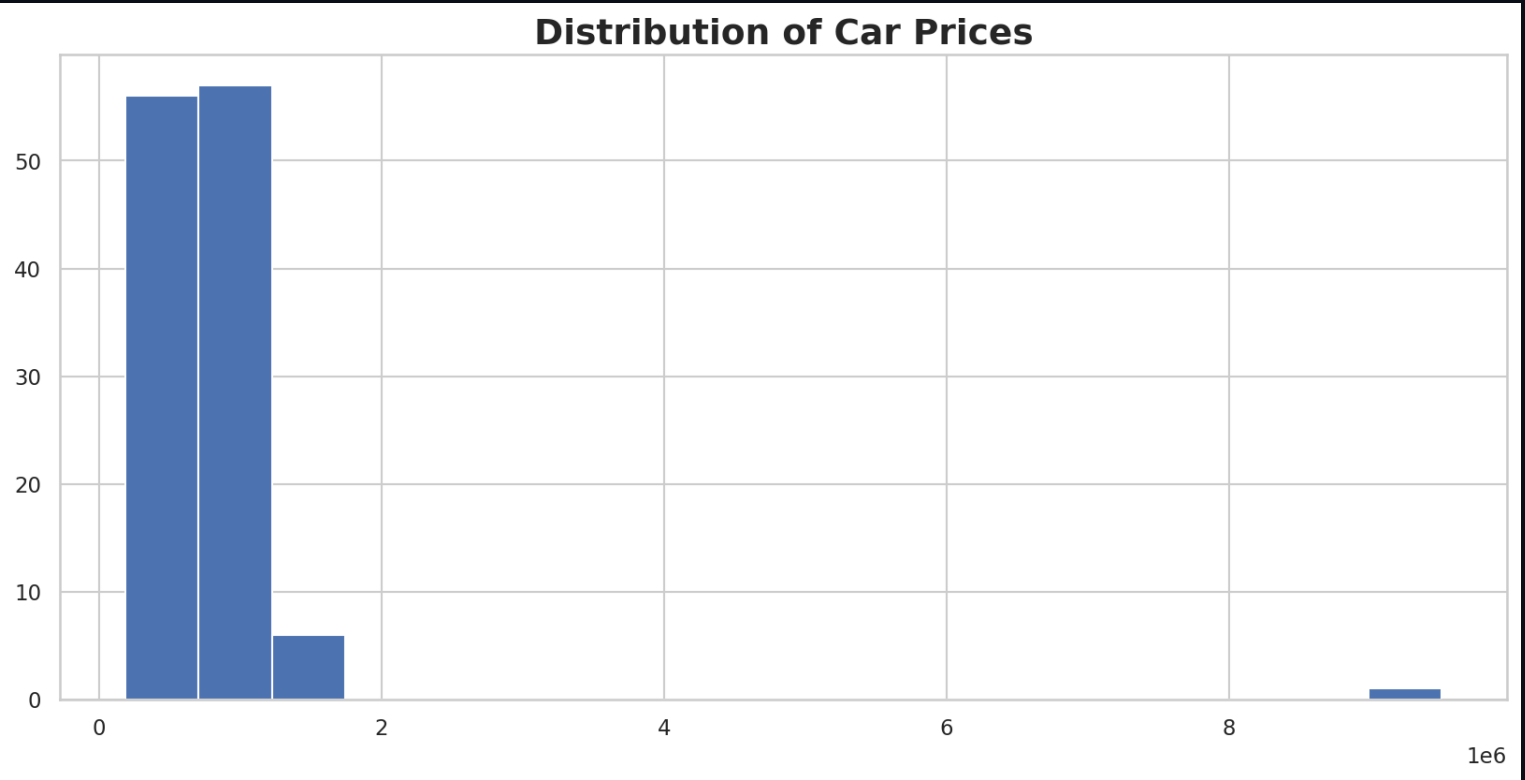
VISUAL EDA • 1

Charts and evidence from the cleaned dataset



Number of Cars by Brand

Observation: Maruti has the largest number of cars in the cleaned dataset.

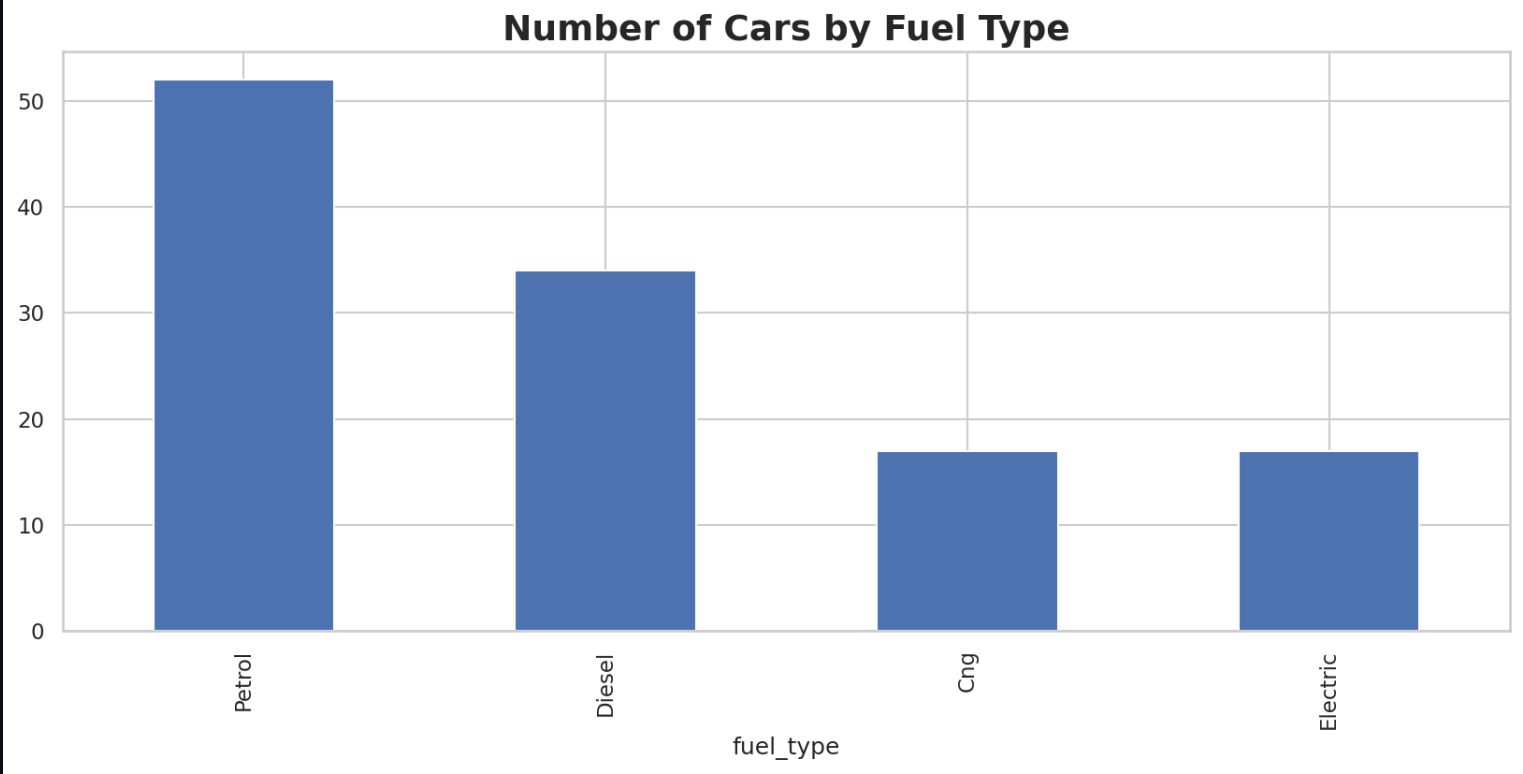


Distribution of Car Prices

Observation: Prices are concentrated toward the lower range, with a high-value outlier; range is ₹189,000 to ₹9,500,000.

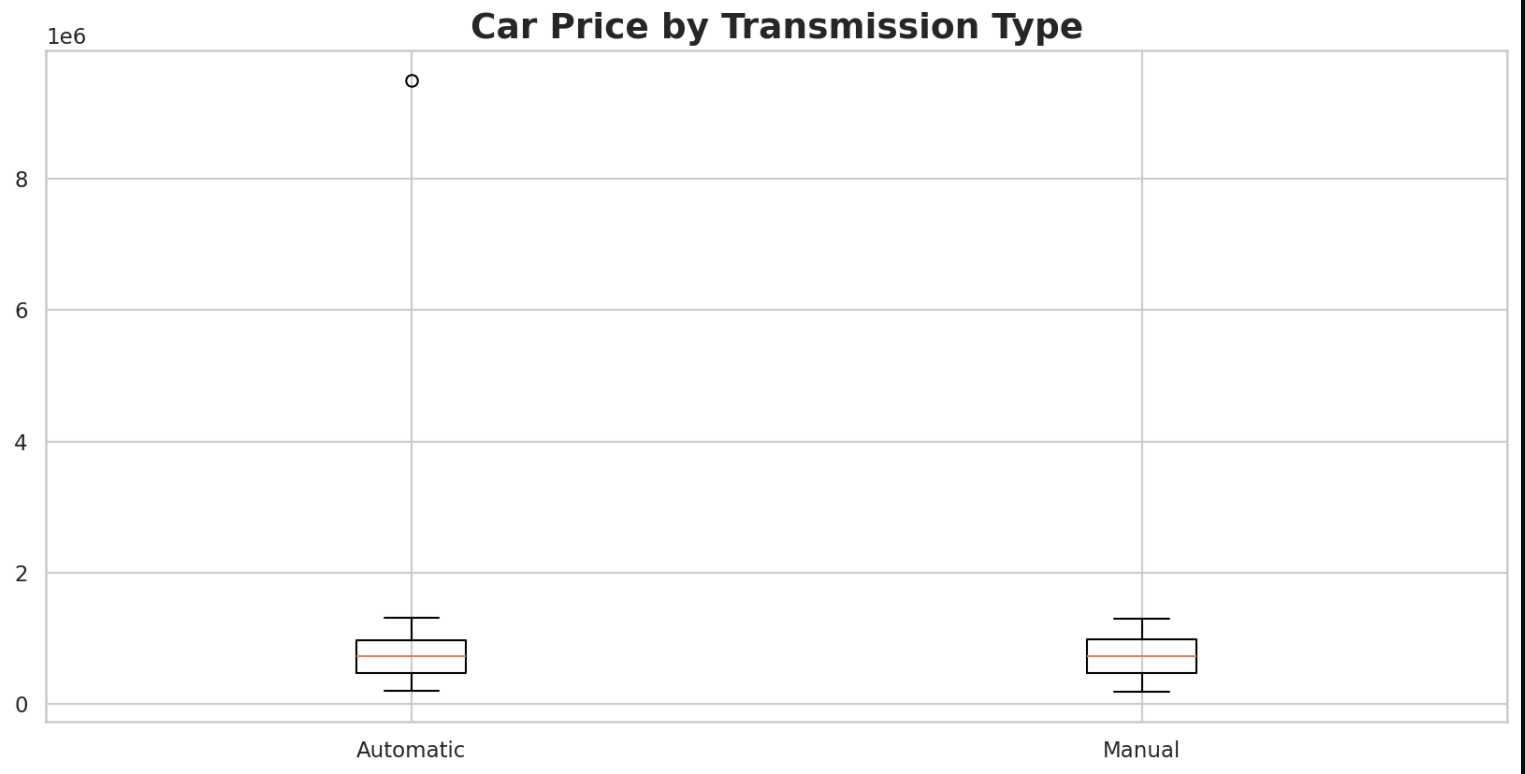
VISUAL EDA • 2

Charts and evidence from the cleaned dataset



Number of Cars by Fuel Type

Observation: Petrol is the most common fuel type with 52 cars.

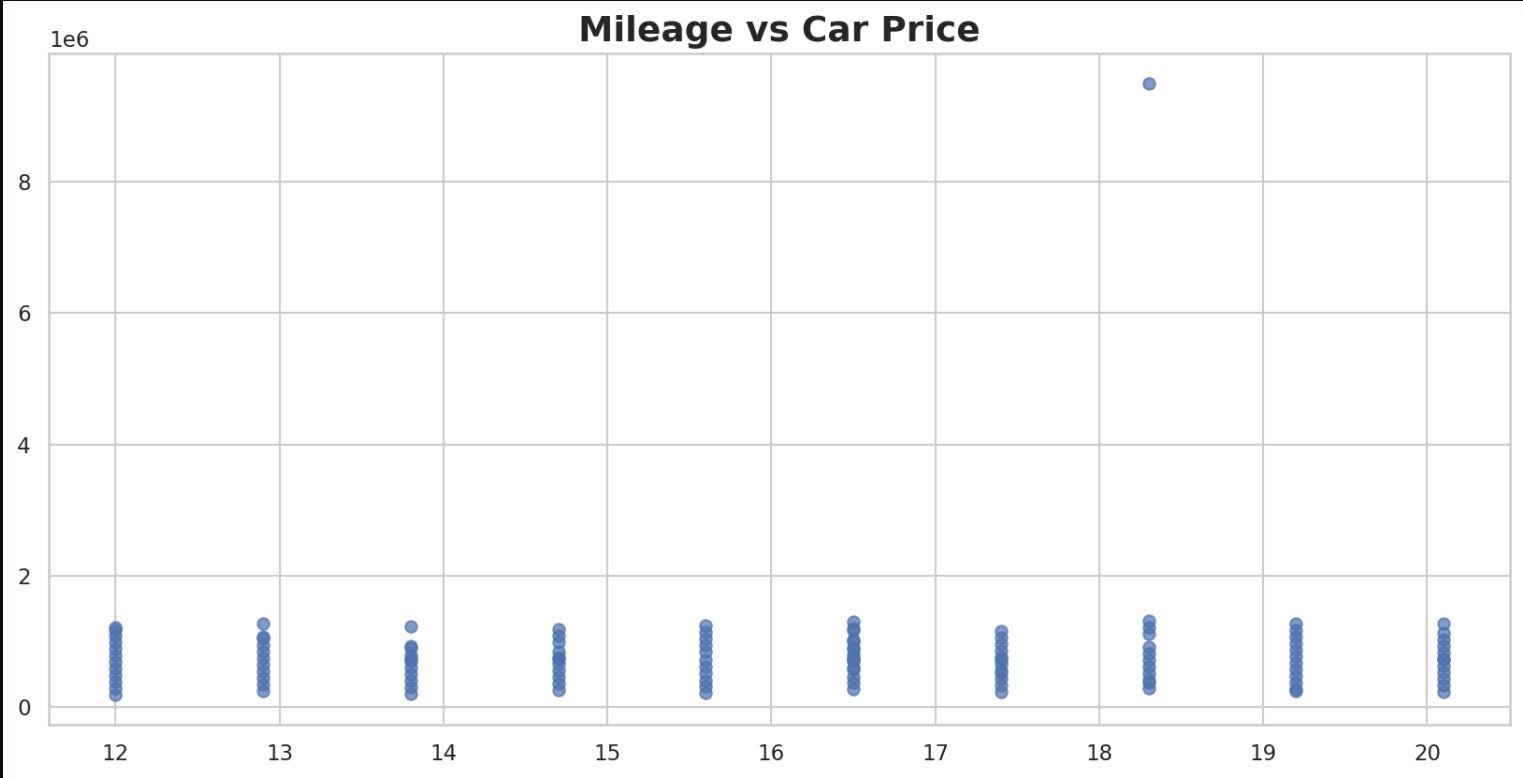


Car Price by Transmission Type

Observation: Automatic cars have a higher average price than manual cars, while the box plot shows substantial price variation.

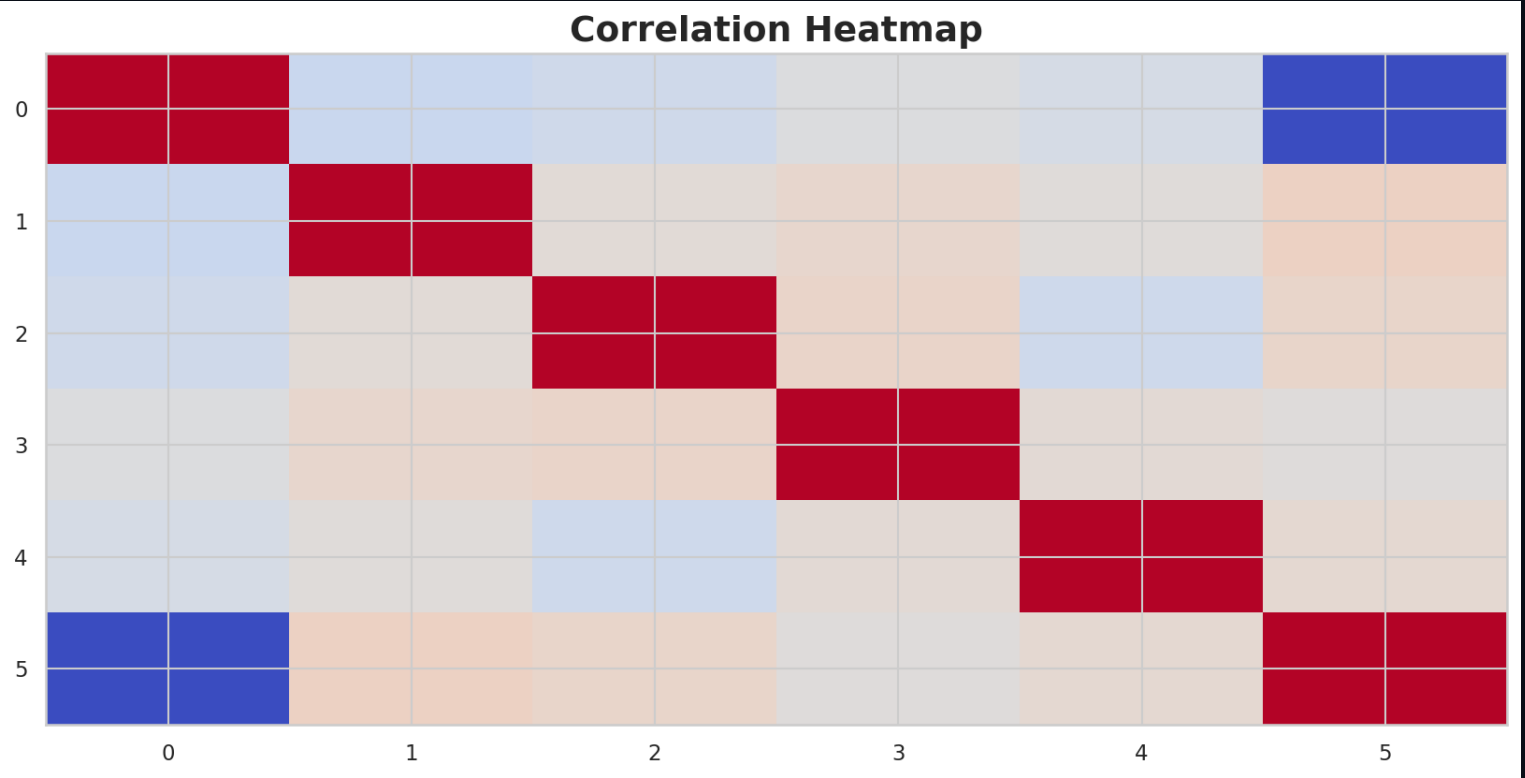
VISUAL EDA • 3

Charts and evidence from the cleaned dataset



Mileage vs Car Price

Observation: Mileage and price show very little linear association in this dataset (correlation ≈ 0.03 for km driven vs price).

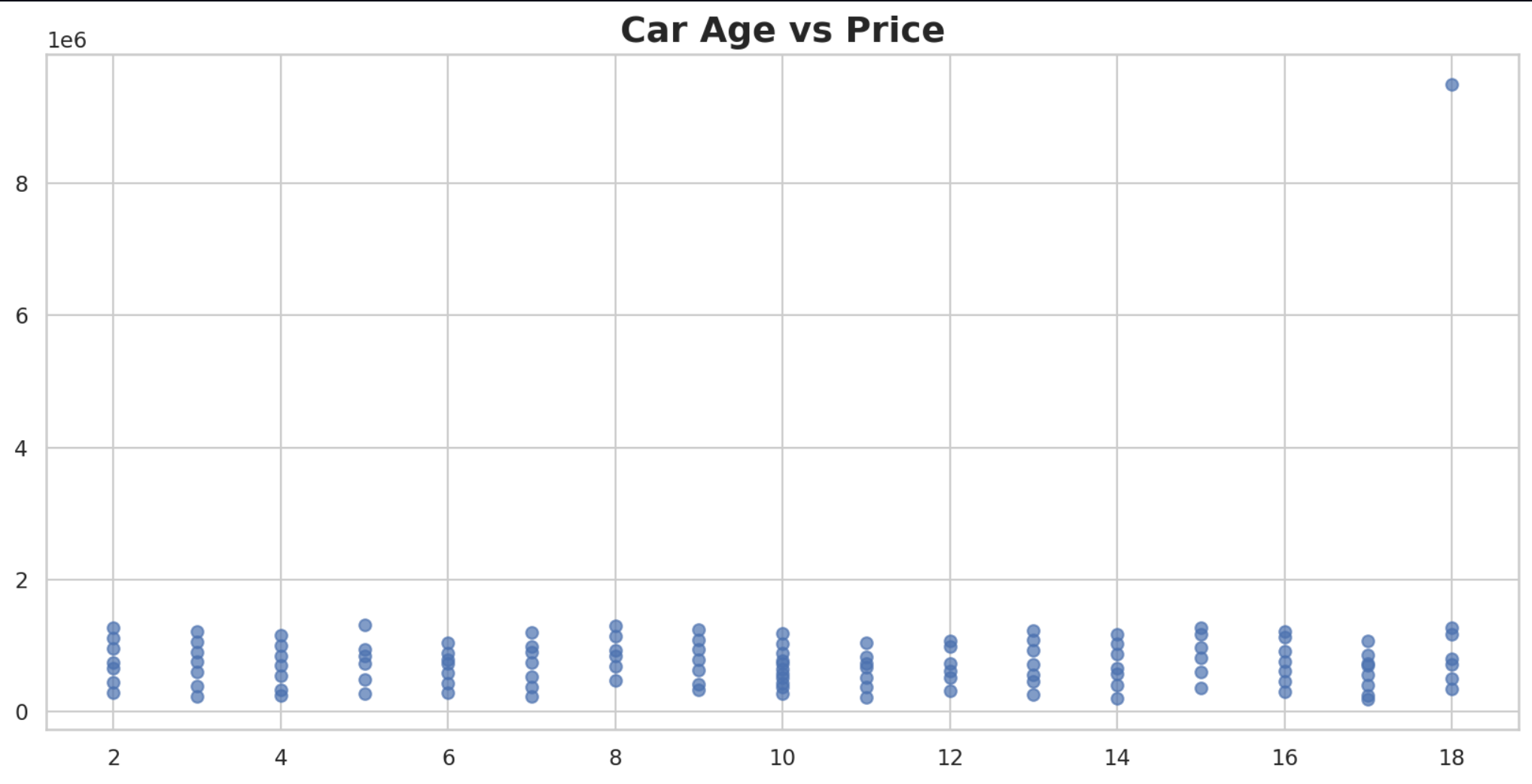


Correlation Heatmap

Observation: The heatmap helps compare linear relationships among the main numerical variables.

VISUAL EDA • 4

Charts and evidence from the cleaned dataset



Car Age vs Price

Observation: Car age has a weak positive linear correlation with price in this cleaned sample ($r \approx 0.13$).

07 | KEY FINDINGS & LIMITATIONS

Evidence-based summary of the EDA

- 1 Maruti is the most common brand, with 35 cars in the cleaned dataset.
- 2 Petrol is the most common fuel type, with 52 cars.
- 3 Average listed price is higher for automatic cars (₹840,506.85) than manual cars (₹733,191.49).
- 4 Prices range from ₹189,000 to ₹9,500,000; the maximum is a strong high-value outlier.
- 5 Car age vs price correlation is 0.13, indicating a weak linear relationship in this sample.
- 6 Kilometres driven vs price correlation is 0.03, indicating almost no linear relationship in this sample.

Limitations: small sample (120 rows), unusual price record, and correlations do not establish causation.

CONCLUSION

This EDA converted a mixed-quality cars dataset into a consistent 120-row analysis table and used multiple visualizations to examine distributions and relationships.

The analysis highlights brand and fuel-type composition, price variation, transmission differences, and weak linear relationships between price and age or kilometres driven.

The workflow demonstrates the core EDA cycle:
understand → clean → analyze → visualize → communicate.

Thank you